

Zongxin Yang, Ph.D.

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in Zongxin Yang

🌐 <https://z-x-yang.github.io/>

Experience

- 2026/08 – Now 📌 **ZJU100 Young Professor.** College of Artificial Intelligence, Zhejiang University. Hangzhou, China.
- 2024/08 – 2026/08 📌 **Research Fellow.** Harvard Medical School, Harvard University. Boston, MA, US.
- 2021/11 – 2024/07 📌 **Postdoctoral Researcher.** CCAI, Zhejiang University. Hangzhou, China.
- 2019/09 – 2021/09 📌 **Research Intern.** Baidu Research. Beijing, China.
Outstanding Intern of 2021.

Education

- 2018/08 – 2021/09 📌 **Ph.D. in Computer Science**
University of Technology Sydney
Advisor: Prof. Yi Yang
Thesis title: *Sequence Modelling with Deep Learning for Visual Content Generation and Understanding.*
- 2014/09 – 2018/06 📌 **B.E. in Electronic and Information Engineering**
University of Science and Technology of China
GPA: 3.99/4.3 Ranking: 3/95 **Guo Moruo Scholarship (Highest Scholarship)**

Research Publications

Preprints and Journal Articles

- 1 K. U, S. M. Zhang, Z. Yu, *et al.*, “AI-powered medicinal chemistry and translational drug development,” *Chemical Society Reviews*, 2026. 🔗 DOI: 10.1039/d5cs01469g.
- 2 Z. Zhang*, **Z. Yang***, A. Liu*, *et al.*, “Learning residue-level context for modeling protein-protein interactions,” *bioRxiv*, 2026. 🔗 DOI: 10.64898/2026.06.01.729118.
- 3 D. Zhou, Y. Li, **Z. Yang**, and Y. Yang, “RefineAnything: Multimodal region-specific refinement for perfect local details,” *arXiv preprint arXiv:2604.06870*, 2026.
- 4 K. F. Greco*, **Z. Yang***, M. Li, *et al.*, “A weakly supervised transformer for rare disease diagnosis and subphenotyping from ehRs with pulmonary case studies,” *npj Digital Medicine*, 2026.
- 5 K. Zhang, **Z. Yang**, M. Han, *et al.*, “Selongvlm: Empowering long video language models with self-corrective clip selection,” *TPAMI*, 2026.
- 6 C. Li, J. Zhang, S. Lin, *et al.*, “Efficient training of large vision models via advanced automated progressive learning,” *TPAMI*, 2026.
- 7 **Z. Yang**, H. Yuan, R. Sayeed, *et al.*, “Clines: Clinical llm-based information extraction and structuring agent,” *medRxiv*, pp. 2025–12, 2025.
- 8 D. Zhou, Y. Li, F. Ma, **Z. Yang***, and Y. Yang, “Migc++: Advanced multi-instance generation controller for image synthesis,” *TPAMI*, 2025.
- 9 Y. Zhu, K. Li, and **Z. Yang**, “Exploiting efficientSAM and temporal coherence for audio-visual segmentation,” *IEEE Transactions on Multimedia*, 2025.

- 10 J. Liu, J. Zhang, S. Yang, *et al.*, “Towards general-purpose video reconstruction through synergy of grid-splicing diffusion and large language models,” *IEEE Transactions on Circuits and Systems for Video Technology*, 2025.
- 11 J. Liu, K. U, D. Rana, *et al.*, “Prompt-based multimodal representation learning for drug repurposing,” *Briefings in Bioinformatics*, vol. 26, no. 6, bba636, 2025.
- 12 **Z. Yang**, X. Wang, J. Miao, Y. Wei, W. Wang, and Y. Yang, “Scalable video object segmentation with identification mechanism,” *TPAMI*, 2024.
- 13 **Z. Yang**, Y. Wei, and Y. Yang, “Collaborative video object segmentation by multi-scale foreground-background integration,” *TPAMI*, 2021.
- 14 J. Ma*, **Z. Yang***, S. Kim, *et al.*, “Medsam2: Segment anything in 3d medical images and videos,” *arXiv preprint arXiv:2504.03600*, 2025.
- 15 Y. Xu, **Z. Yang**, and Y. Yang, “Photorealistic text-to-3d avatar generation with constraints for decoupled geometry and appearance,” *ToMM*, 2025.
- 16 X. Pan, **Z. Yang***, S. Bai, and Y. Yang, “Gd²-nerf: Generative detail compensation via gan and diffusion for one-shot generalizable neural radiance fields,” *ToMM*, 2025.
- 17 J. Liu and **Z. Yang**, “Test-time adaptation for real-world video adverse weather restoration with meta batch normalization,” *IEEE Transactions on Circuits and Systems for Video Technology*, 2025.
- 18 K. Li, T. Jiang, **Z. Yang**, Y. Yang*, Y. Zhuang, and J. Xiao, “Explore synergistic interaction across frames for interactive video object segmentation,” *TCSVT*, 2024.
- 19 C. Liang, L. Zhu, **Z. Yang**, W. Chen, and Y. Yang, “Noise-tolerant hybrid prototypical learning with noisy web data,” *ToMM*, vol. 20, no. 10, pp. 1–19, 2024.
- 20 M. Li, J. Tao, **Z. Yang**, and Y. Yang, “Human101: Training 100+ fps human gaussians in 100s from 1 view,” *Preprint*, 2024.
- 21 Y. Lu, F. Ni, H. Wang, *et al.*, “Show me a video: A large-scale narrated video dataset for coherent story illustration,” *TMM*, 2023.
- 22 S. Jiao, V. Goel, S. Navasardyan, *et al.*, “Collaborative content-dependent modeling: A return to the roots of salient object detection,” *TIP*, 2023.

Conference Proceedings

- 1 J. Ruan, R. Quan, **Z. Yang***, L. Xu, and Y. Yang, “Beyond independent genes: Learning module-inductive representations for gene perturbation prediction,” in *ICML*, 2026.
- 2 Z. Zhang, Z. Chen, **Z. Yang**, and Y. Yang, “Are image-to-video models good zero-shot image editors?” In *CVPR*, 2026.
- 3 W. Song, H. Jiang, **Z. Yang**, R. Quan, and Y. Yang, “Insert anything: Image insertion via in-context editing in dit,” in *AAAI*, 2026.
- 4 R. Zhao, H. Zheng, **Z. Yang**, H. Fan, and Y. Yang, “Stroke3d: Lifting 2d strokes into rigged 3d model via latent diffusion models,” in *ICLR*, 2026.
- 5 D. Zhou, M. Li, **Z. Yang**, *et al.*, “Bidedpo: Conditional image generation with simultaneous text and condition alignment,” in *ICLR*, 2026.
- 6 K. Zhang, **Z. Yang**, M. Han, *et al.*, “Progressive online video understanding with evidence-aligned timing and transparent decisions,” in *ICLR*, 2026.
- 7 K. Li, **Z. Yang***, L. Chen, Y. Yang, and J. Xiao, “Catr: Combinatorial-dependence audio-queried transformer for audio-visual video segmentation,” in *ACM MM (Best Paper Award, Corresponding author)*, 2023.
- 8 **Z. Yang**, G. Chen, X. Li, W. Wang, and Y. Yang, “Doraemongpt: Toward understanding dynamic scenes with large language models (exemplified as a video agent),” in *ICML*, 2024.

- 9 **Z. Yang** and Y. Yang, “Decoupling features in hierarchical propagation for video object segmentation,” in *NeurIPS*, 2022.
- 10 **Z. Yang**, Y. Wei, and Y. Yang, “Associating objects with transformers for video object segmentation,” in *NeurIPS*, 2021.
- 11 **Z. Yang**, X. Yu, and Y. Yang, “Dsc-posenet: Learning 6dof object pose estimation via dual-scale consistency,” in *CVPR*, 2021.
- 12 **Z. Yang**, Y. Wei, and Y. Yang, “Collaborative video object segmentation by foreground-background integration,” in *ECCV*, 2020.
- 13 **Z. Yang**, L. Zhu, Y. Wu, and Y. Yang, “Gated channel transformation for visual recognition,” in *CVPR*, 2020.
- 14 **Z. Yang**, J. Dong, P. Liu, Y. Yang, and S. Yan, “Very long natural scenery image prediction by outpainting,” in *ICCV*, 2019.
- 15 H. Xiong*, **Z. Yang***, J. Yu, *et al.*, “Streaming video understanding and multi-round interaction with memory-enhanced knowledge,” in *ICLR*, 2025.
- 16 D. Zhou*, J. Xie*, **Z. Yang***, and Y. Yang, “3dis: Depth-driven decoupled instance synthesis for text-to-image generation,” in *ICLR*, 2025.
- 17 Z. Zhang, **Z. Yang***, and Y. Yang, “Sifu: Side-view conditioned implicit function for real-world usable clothed human reconstruction,” in *CVPR*, 2024.
- 18 X. Shen, J. Ma, C. Zhou, and **Z. Yang***, “Controllable 3d face generation with conditional style code diffusion,” in *AAAI*, 2024.
- 19 Z. Zhang, J. Xie, Y. Lu, **Z. Yang**, and Y. Yang, “In-context edit: Enabling instructional image editing with in-context generation in large scale diffusion transformer,” in *NeurIPS*, 2025.
- 20 F. Wang, J. Tao, J. Wu, *et al.*, “X-field: A physically grounded representation for 3d x-ray reconstruction,” in *NeurIPS*, 2025.
- 21 S. Gong, Y. Zhuge, L. Zhang, **Z. Yang**, P. Zhang, and H. Lu, “The devil is in temporal token: High quality video reasoning segmentation,” in *CVPR*, 2025, pp. 29 183–29 192.
- 22 D. Zhou, M. Li, **Z. Yang**, and Y. Yang, “Dreamrenderer: Taming multi-instance attribute control in large-scale text-to-image models,” in *ICCV*, 2025.
- 23 Y. Xu, **Z. Yang**, and Y. Yang, “Skdream: Controllable multi-view and 3d generation with arbitrary skeletons,” in *CVPR*, 2025, pp. 314–325.
- 24 J. Xiu, M. Li, **Z. Yang**, W. Ji, Y. Yin, and R. Zimmermann, “Few-shot incremental learning via foreground aggregation and knowledge transfer for audio-visual semantic segmentation,” in *AAAI*, vol. 39, 2025, pp. 8788–8796.
- 25 Z. Zhang, L. Sun, **Z. Yang**, L. Chen, and Y. Yang, “Global-correlated 3d-decoupling transformer for clothed avatar reconstruction,” in *NeurIPS*, 2023.
- 26 X. Pan, **Z. Yang**, J. Ma, C. Zhou, and Y. Yang, “Transhuman: A transformer-based human representation for generalizable neural human rendering,” in *ICCV*, 2023.
- 27 J. Li, **Z. Yang**, X. Wang, J. Ma, C. Zhou, and Y. Yang, “Jotr: 3d joint contrastive learning with transformers for occluded human mesh recovery,” in *ICCV*, 2023.
- 28 Y. Gan, **Z. Yang**, X. Yue, L. Sun, and Y. Yang, “Efficient emotional adaptation for audio-driven talking-head generation,” in *ICCV*, 2023.
- 29 Y. Xu, **Z. Yang**, and Y. Yang, “Integrating boxes and masks: A multi-object framework for unified visual tracking and segmentation,” in *ICCV*, 2023.
- 30 S. Huang, J. Jia, **Z. Yang**, *et al.*, “Shuffled autoregression for motion interpolation,” in *ICASSP*, IEEE, 2023, pp. 1–5.

- 31 X. Shen, **Z. Yang**, X. Wang, J. Ma, C. Zhou, and Y. Yang, “Global-to-local modeling for video-based 3d human pose and shape estimation,” in *CVPR*, 2023.
- 32 J. Miao, **Z. Yang**, L. Fan, and Y. Yang, “Fedseg: Class-heterogeneous federated learning for semantic segmentation,” in *CVPR*, 2023.
- 33 T. Ma, Y. Sun, **Z. Yang**, and Y. Yang, “Prod: Prompting-to-disentangle domain knowledge for cross-domain few-shot image classification,” in *CVPR*, 2023.
- 34 G. Li, Y. Sun, **Z. Yang**, and Y. Yang, “Decompose to generalize: Species-generalized animal pose estimation,” in *ICLR*, 2023.
- 35 D. Zhou, **Z. Yang**, and Y. Yang, “Pyramid diffusion models for low-light image enhancement,” in *IJCAI*, 2023.
- 36 Y. Xu, **Z. Yang**, and Y. Yang, “Video object segmentation in panoptic wild scenes,” in *IJCAI*, 2023.
- 37 S. Huang, **Z. Yang**, L. Li, Y. Yang, and J. Jia, “Avatarfusion: Zero-shot generation of clothing-decoupled 3d avatars using 2d diffusion,” in *ACM MM*, 2023.
- 38 F. Zhu, **Z. Yang**, X. Yu, Y. Yang, and Y. Wei, “Instance as identity: A generic online paradigm for video instance segmentation,” in *ECCV*, 2022.
- 39 M. Kristan, A. Leonardis, J. Matas, *et al.*, “The tenth visual object tracking vot2022 challenge results,” in *ECCV*, 2022.
- 40 Y. Xu, Y. Sun, **Z. Yang**, J. Miao, and Y. Yang, “H2fa r-cnn: Holistic and hierarchical feature alignment for cross-domain weakly supervised object detection,” in *CVPR*, 2022.
- 41 X. Pan, P. Li, **Z. Yang**, *et al.*, “In-n-out generative learning for dense unsupervised video segmentation,” in *ACM MM*, 2022.

Technical Reports

- 1 Y. Cheng, L. Li, Y. Xu, *et al.*, “Segment and track anything,” in github.com/z-x-yang/Segment-and-Track-Anything (**Project leader, 2.4k stars**), 2023.
- 2 **Z. Yang**, P. Li, Q. Feng, Y. Wei, and Y. Yang, “Going deeper into embedding learning for video object segmentation,” in *ICCV Workshops*, 2019.
- 3 **Z. Yang**, Y. Ding, Y. Wei, and Y. Yang, “Cfbi+: Collaborative video object segmentation by multi-scale foreground-background integration,” in *CVPR Workshops*, 2020.
- 4 **Z. Yang**, J. Zhang, W. Wang, *et al.*, “Towards multi-object association from foreground-background integration,” in *CVPR Workshops*, 2021.
- 5 Y. Xu, J. Li, **Z. Yang**, Y. Yang, and Y. Zhuang, “Zju reler submission for epic-kitchen challenge 2023: Trek-150 single object tracking,” in *CVPR Workshops*, 2023.
- 6 J. Li, Y. Xu, **Z. Yang**, Y. Yang, and Y. Zhuang, “Zju reler submission for epic-kitchen challenge 2023: Semi-supervised video object segmentation,” in *CVPR Workshops*, 2023.
- 7 W. Wang, Y. Sun, **Z. Yang**, and Y. Yang, “Leveraging vision and vision-language models into large-scale product retrieval,” in *CVPR Workshops*, 2022.
- 8 C. Liang, Y. Wu, T. Zhou, *et al.*, “Rethinking cross-modal interaction from a top-down perspective for referring video object segmentation,” in *CVPR Workshops*, 2021.
- 9 C. Liang, **Z. Yang**, J. Miao, Y. Wei, and Y. Yang, “Memory aggregated cfbi+ for interactive video object segmentation,” in *CVPR Workshops*, 2020.
- 10 Q. Feng, **Z. Yang**, P. Li, Y. Wei, and Y. Yang, “Dual embedding learning for video instance segmentation,” in *ICCV Workshops*, 2019.

Awards (Selected)

- 2023
 - [ACM MM 2023] **Best Paper Award.**
 - [ICCV 2023] **1st** in the VOTS 2023 challenge.
 - [CVPR 2023] **1st** in Semi-Supervised Video Object Segmentation of EPIC-Kitchens Challenges.
 - [CVPR 2023] **1st** in TREK-150 Object Tracking of EPIC-Kitchens Challenges.
- 2022
 - [ECCV 2022] **1st** in the VOT 2022 real-time segmentation tracking challenge.
 - [ECCV 2022] **1st** in the VOT 2022 short-term segmentation tracking challenge.
 - [CVPR 2022] **1st** in eBay eProduct Visual Search Challenge.
- 2021
 - [CVPR 2021] **1st** in the 3rd Large-scale Video Object Segmentation Challenge.
 - [CVPR 2021] **1st** in the 3rd Large-scale Referring Video Object Segmentation Challenge.
 - **Outstanding Intern.** Awarded by Baidu Inc.
- 2018
 - **Guo Moruo Scholarship.** Awarded by USTC (**Highest Scholarship, Top 1.5%**).
 - Outstanding Graduates. Awarded by USTC.

Invited Talks and Lectures (Selected)

- 2026
 - **Biomedical AI for Rare Disease: How AI Learns from Electronic Health Records**
Guest lecture, Harvard Medical School Summer Lecture, Boston, MA, USA, Jul 2026
- 2025
 - **Cross-modal Controllable Generation by Post-training**
VALSE Webinar, China, Jun 2025
 - **Cross-modal Controllable Image and 3D Generation**
Shanghai AI Lab, Shanghai, China, Jan 2025
- 2024
 - **Object-centric Intelligence in Dynamic Scenes**
Beijing Jiaotong University, China, Jun 2024
- 2023
 - **Decoupled Memory of Associating Objects with Transformers**
1st at the VOTS2023 challenge, ICCV 2023, Paris, France, Oct 2023
 - **Segment and Track Anything**
VALSE Webinar, China, Aug 2023
 - **Digital Human Generation and Reconstruction Driven by Knowledge and Data**
Zhejiang University, China, Mar 2023

Invited Talks and Lectures (Selected) (continued)

- 2022  **Decoupling Features in Hierarchical Propagation for Video Object Segmentation**
Spotlight presentation, NeurIPS 2022, New Orleans, USA, Nov 2022
-  **Associating Objects with Multi-scale Transformers for Video Object Segmentation**
Winner presentation, VOT2022 Workshop, ECCV 2022, Tel-Aviv, Israel, Oct 2022
-  **Associating Multi-object Segmentation in Video Understanding**
VALSE Webinar, China, Mar 2022
- 2021  **Towards Multi-Object Association from Foreground-Background Integration**
1st at the 3rd Large-scale Video Object Segmentation Challenge, CVPR 2021, Online, June 2021
-  **Associating Objects with Transformers for Video Object Segmentation**
Tutorial: Data-Efficient Learning in An Imperfect World, CVPR 2021, Online, June 2021
- 2020  **Collaborative Video Object Segmentation by Foreground-background Integration**
Spotlight presentation, ECCV 2020, Online, Aug 2020
-  **Collaborative Video Object Segmentation by Multi-Scale Foreground-Background Integration**
2nd at the 2020 DAVIS Challenge, CVPR 2020, Online, Jun 2020
- 2019  **Going Deeper into Embedding Learning for Video Object Segmentation**
The 2nd Large-scale Video Object Segmentation Challenge, ICCV 2019, Seoul, Korea, Oct 2019

Professional Services

- Area Chair  **ECCV 2026**
CVPR 2026 (Outstanding Area Chair)
- Workshop Organizer  **The 5th Pixel-level Video Understanding in the Wild Challenge** at CVPR 2026.
The 4th Pixel-level Video Understanding in the Wild Challenge at CVPR 2025.
The 3rd Pixel-level Video Understanding in the Wild Challenge at CVPR 2024.
The 2nd Pixel-level Video Understanding in the Wild Challenge at CVPR 2023.
- Journal Reviewer  **TPAMI, IJCV, TIP, TCSVT, TIST, KBS, FCS, etc.**
- Conference Reviewer  **NeurIPS, ICML, CVPR, ICCV, SIGGRAPH Asia, AACL, IJCAI, etc.**
Outstanding Reviewer of ICCV 2023.